C++ Practice Questions: Functions

CSE141 Introduction to Programming

Instructions

- Write a separate C++ program for each question.
- Use only the concepts covered in lectures and labs.
- Read each question carefully and identify the required function parameters, return type, and function calls.
- Unless otherwise stated, take input in `main()` and pass the required values to functions.
- For pointer questions, pass valid addresses and do not dereference invalid or uninitialized pointers.
- Questions are divided into Easy, Medium, and Hard levels. Each topic contains a mixture of difficulty levels.

Topic 1: Function Basics

Question 1 ***[Easy]***

Write a function named `greet` that displays `Welcome to C++ Functions!`. Call the function from `main()`.

Question 2 ***[Easy]***

Write a function named `showLine` that displays a line of 30 asterisks. Call it three times to create a simple heading.

Question 3 ***[Medium]***

Write a function that displays a student's name, roll number, and semester. Call the function from `main()`.

Question 4 ***[Medium]***

Create a function named `displayMenu` that displays a menu containing three options. The function should not take any arguments.

Question 5 ***[Hard]***

Write a program with separate functions to display a program title, instructions, and a closing message. Call them in the correct order from `main()`.

Topic 2: Parameters and Arguments

Question 6 ***[Easy]***

Write a function named `printNumber` that accepts one integer parameter and displays it. Take the number from the user.

Question 7 ***[Easy]***

Create a function that accepts two integers and displays both values.

Question 8 ***[Medium]***

Write a function that accepts the length and width of a rectangle and displays its area and perimeter.

Question 9 ***[Medium]***

Create a function that accepts a student's name and marks, then displays them in a formatted result statement.

Question 10 ***[Hard]***

Write a function that accepts three numbers and displays them in ascending order. Do not take input inside the function.

Topic 3: Return Values

Question 11 ***[Easy]***

Write a function named `getSquare` that accepts an integer and returns its square. Display the returned value in `main()`.

Question 12 ***[Easy]***

Create a function that accepts two integers and returns their sum.

Question 13 ***[Medium]***

Write a function that accepts the radius of a circle and returns its area. Display the result with two decimal places.

Question 14 ***[Medium]***

Create a function that accepts marks and returns `true` if the student has passed (50 or above), otherwise `false`.

Question 15 ***[Hard]***

Write separate functions to calculate the subtotal, discount amount, and final bill. The main function should display all three returned values.

Topic 4: Void Functions vs. Value-Returning Functions

Question 16 ***[Easy]***

Write a `void` function that accepts a name and displays a greeting.

Question 17 ***[Easy]***

Write a value-returning function that accepts an integer and returns whether it is even or odd as a suitable result.

Question 18 ***[Medium]***

Create two functions: one `void` function to display a student's details and one returning function to calculate their percentage.

Question 19 ***[Medium]***

Write a `void` function that accepts two numbers and displays their average. Then rewrite it as a function that returns the average.

Question 20 ***[Hard]***

Design a program using a `void` function to display a menu and returning functions to calculate the area of a circle, rectangle, or triangle according to the user's choice.

Topic 5: Pass by Value

Question 21 ***[Easy]***

Write a function that accepts an integer by value, changes the parameter by adding 10, and displays it. Display the original variable in `main()` after the call.

Question 22 ***[Easy]***

Create a function that accepts a `double` by value and multiplies the parameter by 2. Check whether the original variable changes.

Question 23 ***[Medium]***

Write a function that accepts two integers by value and swaps the parameter values. Display the variables before and after calling the function.

Question 24 ***[Medium]***

Create a function that accepts a student's marks by value and adds 5 bonus marks to its local parameter. Display the value inside and outside the function.

Question 25 ***[Hard]***

Write a program with two functions that each receive the same integer by value and modify their local copies differently. Display the original value after both calls and explain the result.

Topic 6: Pass by Reference

Question 26 ***[Easy]***

Write a function that accepts an integer by reference and increases the original variable by 1.

Question 27 ***[Easy]***

Create a function that accepts a `double` by reference and converts the value from dollars to cents.

Question 28 ***[Medium]***

Write a function that swaps two integers using reference parameters. Display the values before and after the function call.

Question 29 ***[Medium]***

Create a function that accepts a student's marks by reference and adds 5 bonus marks to the original variable.

Question 30 ***[Hard]***

Write a function that accepts three integer variables by reference and rearranges them so that the first contains the smallest value and the third contains the largest value.

Topic 7: Pointers as Function Parameters

Question 31 ***[Easy]***

Write a function that accepts an integer pointer and displays the value stored at the pointed-to address. Call it by passing the address of an integer.

Question 32 ***[Easy]***

Create a function that accepts an integer pointer and changes the pointed-to value to 100.

Question 33 ***[Medium]***

Write a function that accepts two integer pointers and swaps the values they point to.

Question 34 ***[Medium]***

Create a function that accepts a pointer to a `double` and increases the pointed-to value by 10

Question 35 ***[Hard]***

Write a function that accepts pointers to three integers and stores the largest value in the variable pointed to by the third pointer. Do not use global variables.

Topic 8: Default Parameters

Question 36 ***[Easy]***

Write a function named `showMessage` that accepts a message and a repetition count. Give the repetition count a default value of 1.

Question 37 ***[Easy]***

Create a function that calculates the power of a number, with the exponent defaulting to 2.

Question 38 ***[Medium]***

Write a function that calculates the total price using quantity and unit price. Set the quantity's default value to 1.

Question 39 ***[Medium]***

Create a function that displays a student's name and department. Give the department parameter a default value of `Computer Science`.

Question 40 ***[Hard]***

Write a function to calculate simple interest using principal, rate, and time. Give rate and time default values. Call the function using all arguments, two arguments, and only the principal; ensure the defaults make each call valid.

Topic 9: Recursive Functions

Question 41 ***[Easy]***

Write a recursive function that displays the numbers from 1 to N. Take N from the user.

Question 42 ***[Easy]***

Write a recursive function that calculates the sum of integers from 1 to N.

Question 43 ***[Medium]***

Write a recursive function to calculate the factorial of a non-negative integer.

Question 44 ***[Medium]***

Write a recursive function that calculates the sum of the digits of a positive integer.

Question 45 ***[Hard]***

Write a recursive function that determines whether a number is a palindrome. Do not use loops or convert the number to a string.

Topic 10: Function Prototypes, Overloading, and Scope

Question 46 ***[Easy]***

Write a function prototype for a function named `calculateTotal` that accepts two `double` parameters and returns a `double`. Then define and call it.

Question 47 ***[Easy]***

Declare a function prototype above `main()`, define the function below `main()`, and call it from `main()`.

Question 48 ***[Medium]***

Overload a function named `area` to calculate the area of a square, rectangle, and circle using suitable parameter lists.

Question 49 ***[Medium]***

Create a program with a local variable inside `main()` and another local variable with the same name inside a function. Display both and observe their scope.

Question 50 ***[Hard]***

Overload a function named `calculate` to perform addition for two integers, two doubles, and three integers. Call each version and display the results.

Mixed Practice

These questions combine two or more function concepts covered above.

Question 51 ***[Easy]***

Write a function that accepts two integers by value and returns their product. Take input in `main()` and display the returned result.

Question 52 ***[Easy]***

Create a `void` function with a default parameter that displays a welcome message. Call it once with no argument and once with a custom message.

Question 53 ***[Medium]***

Write a function that accepts an integer by reference, checks whether it is negative, and changes it to its positive value. Display the variable before and after the call.

Question 54 ***[Medium]***

Create a recursive function to calculate the sum of even numbers from 2 up to N. Take N from the user and display the returned sum.

Question 55 ***[Medium]***

Write a function that accepts an integer pointer and a second integer. Update the pointed-to variable by adding the second integer to it. Display the result in `main()`.

Question 56 ***[Hard]***

Write a function named `updateResult` that accepts marks by reference and a bonus by value. Add the bonus to the original marks, but do not allow the final marks to exceed 100. Display the result in `main()`.

Question 57 ***[Hard]***

Create overloaded functions named `maximum` for two integers and three doubles. Use a function prototype for each overload and call both from `main()`.

Question 58 ***[Hard]***

Write a recursive function that counts how many times a digit occurs in a positive integer. The number and digit should be passed as arguments; return the count.